Project: Advanced News Recommender System

1) Install & Import *Libraries*

```
# Install necessary packages
!pip install pandas numpy scikit-learn nltk textblob

# Import libraries
import pandas as pd
import numpy as np
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.metrics.pairwise import cosine_similarity
from textblob import TextBlob
import nltk
from nltk.corpus import stopwords
import re
nltk.download('stopwords')
```

```
Requirement already satisfied: pandas in /usr/local/lib/python3.12/dist-packages (2.2.2)
Requirement already satisfied: numpy in /usr/local/lib/python3.12/dist-packages (2.0.2)
Requirement already satisfied: scikit-learn in /usr/local/lib/python3.12/dist-packages (1.6.1)
Requirement already satisfied: nltk in /usr/local/lib/python3.12/dist-packages (3.9.1)
Requirement already satisfied: textblob in /usr/local/lib/python3.12/dist-packages (0.19.0)
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.12/dist-packages (from pandas) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas) (2025.2)
Requirement already satisfied: scipy>=1.6.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn) (1.16.2)
Requirement already satisfied: joblib>=1.2.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn) (1.5.2)
Requirement already satisfied: threadpoolctl>=3.1.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn) (3.6.0)
Requirement already satisfied: click in /usr/local/lib/python3.12/dist-packages (from nltk) (8.3.0)
Requirement already satisfied: regex>=2021.8.3 in /usr/local/lib/python3.12/dist-packages (from nltk) (2024.11.6)
Requirement already satisfied: tqdm in /usr/local/lib/python3.12/dist-packages (from nltk) (4.67.1)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.8.2->pandas) (1.17
[nltk_data] Downloading package stopwords to /root/nltk_data...
[nltk_data]   Unzipping corpora/stopwords.zip.
True
```

2) Sample Dataset

```
data = {
    'id': [1, 2, 3, 4, 5, 6],
    'title': [
        'AI transforming healthcare industry',
        'Stock markets reach new highs',
        'New breakthrough in cancer research',
        'Economic growth slows amid inflation',
        'AI used to detect diseases earlier',
        'Technology companies report record profits'
    ],
    'content': [
        'Artificial intelligence is playing a key role in transforming the healthcare industry by
        'Global stock markets hit record highs as investors remain optimistic about the technology
        'Scientists have made a major breakthrough in cancer research, discovering new genetic mar
        'Economic growth has slowed in several countries as rising inflation and interest rates be
        'AI systems are being deployed to detect diseases earlier than before, enabling doctors to
        'Major technology companies report record profits due to increased demand for cloud comput
    ]
}

df = pd.DataFrame(data)
print(df[['id', 'title']])
```

```
   id                                   title
0   1        AI transforming healthcare industry
1   2              Stock markets reach new highs
2   3        New breakthrough in cancer research
3   4       Economic growth slows amid inflation
4   5         AI used to detect diseases earlier
5   6  Technology companies report record profits
```

3) Text Preprocessing

```
stop_words = set(stopwords.words('english'))

def clean_text(text):
    # Lowercase
    text = text.lower()
    # Remove special characters and numbers
    text = re.sub(r'[^a-zA-Z\s]', '', text)
    # Remove stopwords
    tokens = text.split()
    tokens = [word for word in tokens if word not in stop_words]
    return ' '.join(tokens)

df['clean_text'] = df['title'] + ' ' + df['content']
df['clean_text'] = df['clean_text'].apply(clean_text)
print(df[['title', 'clean_text']])
```

```
                                        title  \
0         AI transforming healthcare industry
1               Stock markets reach new highs
2         New breakthrough in cancer research
3        Economic growth slows amid inflation
4          AI used to detect diseases earlier
5  Technology companies report record profits

                                          clean_text
0  ai transforming healthcare industry artificial...
1  stock markets reach new highs global stock mar...
2  new breakthrough cancer research scientists ma...
3  economic growth slows amid inflation economic ...
4  ai used detect diseases earlier ai systems dep...
5  technology companies report record profits maj...
```

4) TF-IDF Vectorization

```
vectorizer = TfidfVectorizer(max_features=5000)
tfidf_matrix = vectorizer.fit_transform(df['clean_text'])
```

5) Compute Cosine Similarity

```
cosine_sim = cosine_similarity(tfidf_matrix, tfidf_matrix)
similarity_df = pd.DataFrame(cosine_sim, index=df['title'], columns=df['title'])
print(similarity_df.round(2))
```

```
title                                       AI transforming healthcare industry  \
title
AI transforming healthcare industry                                        1.00
Stock markets reach new highs                                              0.00
New breakthrough in cancer research                                        0.00
Economic growth slows amid inflation                                       0.00
AI used to detect diseases earlier                                         0.09
Technology companies report record profits                                 0.00

title                                       Stock markets reach new highs  \
title
AI transforming healthcare industry                                  0.00
```

```
Stock markets reach new highs                                 1.00
New breakthrough in cancer research                           0.06
Economic growth slows amid inflation                          0.06
AI used to detect diseases earlier                            0.00
Technology companies report record profits                    0.11

title                                        New breakthrough in cancer research  \
title
AI transforming healthcare industry                                  0.00
Stock markets reach new highs                                        0.06
New breakthrough in cancer research                                  1.00
Economic growth slows amid inflation                                 0.00
AI used to detect diseases earlier                                   0.00
Technology companies report record profits                           0.03

title                                        Economic growth slows amid inflation  \
title
AI transforming healthcare industry                                   0.00
Stock markets reach new highs                                         0.06
New breakthrough in cancer research                                   0.00
Economic growth slows amid inflation                                  1.00
AI used to detect diseases earlier                                    0.00
Technology companies report record profits                            0.00

title                                        AI used to detect diseases earlier  \
title
AI transforming healthcare industry                                 0.09
Stock markets reach new highs                                       0.00
New breakthrough in cancer research                                 0.00
Economic growth slows amid inflation                                0.00
AI used to detect diseases earlier                                  1.00
Technology companies report record profits                          0.00

title                                        Technology companies report record profits
title
AI transforming healthcare industry                                         0.00
Stock markets reach new highs                                               0.11
New breakthrough in cancer research                                         0.03
Economic growth slows amid inflation                                        0.00
AI used to detect diseases earlier                                          0.00
Technology companies report record profits                                  1.00
```

6) Recommendation Function

```python
def recommend_articles(title, top_n=3):
    if title not in similarity_df.columns:
        return "Article not found."

    scores = similarity_df[title].sort_values(ascending=False)
    recommended = scores.iloc[1:top_n+1].index.tolist()
    return recommended
```

7) Sentiment Analysis of Articles

```python
def analyze_sentiment(text):
    blob = TextBlob(text)
    polarity = blob.sentiment.polarity
    if polarity > 0:
        return 'Positive'
    elif polarity < 0:
        return 'Negative'
    else:
        return 'Neutral'

df['sentiment'] = df['content'].apply(analyze_sentiment)
print(df[['title', 'sentiment']])
```

```
                                        title sentiment
0         AI transforming healthcare industry  Negative
1               Stock markets reach new highs   Neutral
2         New breakthrough in cancer research  Positive
3        Economic growth slows amid inflation  Positive
4          AI used to detect diseases earlier  Positive
5  Technology companies report record profits  Negative
```

8) Keyword Extraction

```
def extract_keywords(text, n=5):
    vectorizer = TfidfVectorizer(stop_words='english')
    tfidf = vectorizer.fit_transform([text])
    feature_array = np.array(vectorizer.get_feature_names_out())
    tfidf_sorting = np.argsort(tfidf.toarray()).flatten()[::-1]
    top_n_words = feature_array[tfidf_sorting][:n]
    return top_n_words

df['keywords'] = df['clean_text'].apply(lambda x: extract_keywords(x, n=5))
print(df[['title', 'keywords']])
```

```
                                     title  \
0        AI transforming healthcare industry
1              Stock markets reach new highs
2        New breakthrough in cancer research
3       Economic growth slows amid inflation
4         AI used to detect diseases earlier
5  Technology companies report record profits

                                       keywords
0  [transforming, healthcare, industry, treatment...
1            [stock, highs, markets, remain, record]
2  [new, research, cancer, breakthrough, treatments]
3      [economic, growth, inflation, rising, slowed]
4              [earlier, detect, ai, diseases, used]
5   [technology, report, profits, record, companies]
```

9) Advanced Recommendation with Sentiment Filtering

```
def recommend_positive_articles(title, top_n=3):
    similar_titles = recommend_articles(title, top_n=top_n*2)  # Get more to filter
    positive_articles = df[df['title'].isin(similar_titles) & (df['sentiment']=='Positive')]['tit
    return positive_articles[:top_n]

# Sample user input for demonstration
user_input = "AI transforming healthcare industry"

print("\nPositive Recommendations:")
print(recommend_positive_articles(user_input, top_n=3))
```

```
Positive Recommendations:
['New breakthrough in cancer research', 'Economic growth slows amid inflation', 'AI used to detect diseases earlier']
```

10) Full Execution Example

```
print("\n--- User Input ---")
print(user_input)

print("\n--- Top Similar Articles ---")
print(recommend_articles(user_input, top_n=3))

print("\n--- Positive Sentiment Recommendations ---")
print(recommend_positive_articles(user_input, top_n=3))

print("\n--- Keywords in User Article ---")
print(extract_keywords(df[df['title']==user_input]['clean_text'].values[0], n=5))
```

```
--- User Input ---
AI transforming healthcare industry

--- Top Similar Articles ---
['AI used to detect diseases earlier', 'Stock markets reach new highs', 'New breakthrough in cancer research']
```

```
--- Positive Sentiment Recommendations ---
['New breakthrough in cancer research', 'Economic growth slows amid inflation', 'AI used to detect diseases earlier']

--- Keywords in User Article ---
['transforming' 'healthcare' 'industry' 'treatment' 'role']
```